

PROJECT NABEK

COMPLETE DOCUMENT

Founding Preamble + Synthetic Document

Founding Preamble — The first condition · Philosophy · Wampum · Abolition of ownership
Synthetic Document — Vision · Model · Value Chains · Projections · Directory

*"What we offer you is not a job, not an investment, not land to buy."
"It is a responsibility to take on."*

KEY REFERENCES
• Unceded Land — Turtle Island — Year 410 of the 1616 Treaty
• Developed by Nabek, in accord with the Kanienkehaka Longhouse
• Personal project in Quebec Without subsidy Outside government framework
• All amounts in Canadian dollars (CAD) — Exchange rate May 2026: 1 USD = 1.37 CAD
• Real, punctual, Quebec-specific data
• Complete document — June 2026

FOUNDING PREAMBLE

This text is non-negotiable. It is the first condition.

I. THE FOUNDATION — NOT WORDS, ACTIONS

Words fly away. Actions remain. This project does not rest on speeches, declarations of intent, or verbal promises. It rests on concrete actions, taken, repeated, verified.

Whoever joins this project does not sign a contract. They take action. An act of respect toward the peoples who were here before us. An act of care toward the land. An act of renouncing ownership.

Walk the walk, or don't walk at all. If not, there is no place here.

II. THE TWO-ROW WAMPUM

In Year 410 of the 1616 Treaty, the Two-Row Wampum remains in force. It is not a contract signed with the King of France. It is a pact of coexistence. The Kanienkehaka came to us with this Wampum to say: 'Respect the law of this land, or leave — as the Vikings did before you, whose remnants in Newfoundland testify.'

When the project sets foot on Innu land, it reaches an agreement with the Innu. On Mi'kmaq land, with the Mi'kmaq. On Abenaki land, with the Abenaki. Whatever the people, the agreement is made with them. Not with Quebec. Not with Ottawa. Not with a ministry. Not with a notary. Not with a land registry.

This project is outside colonial jurisdiction. And so are you, if you join.

III. THE LAND BELONGS TO NO ONE

Before colonization, this land belonged to no one. It was not 'owned' by anyone. It was inhabited. There was harmony among all forms of life — humans, animals, plants, mycelium, waters, stones. The concept of land ownership did not exist. That is why the land was never ceded. One does not cede what one does not possess.

The Nabek project rests on this fact: **land ownership is a historical error that must be corrected.** Not by transferring ownership from one people to another, but by abolishing the very act of ownership.

IV. THE ABOLITION OF LAND OWNERSHIP

The lands of the Nabek project will never be sold. Never taxed. Never transferred by sale, inheritance, or cession. They are not assets. They are responsibilities.

When the one who tends a plot stops — through exhaustion, death, or choice — it is not an heir who takes over. It is another caretaker, chosen by the community, trained by the cooperative, recognized by the peoples of the territory. The land passes from one living hand to another living hand. Never from one signature to another signature.

The lands belong neither to Quebec. Nor to Indigenous peoples. Nor to the project. **They belong to the living.** We are only temporary caretakers.

If you seek to possess, to capitalize, to pass on a legacy to your children in the form of a land title — this project is not for you. You may close this document now. There is no judgment. But there is no place either.

V. THE COMPLEMENTARITY OF PEOPLES

Two canoes on a river. In each canoe, its own law, its own language, its own spirituality. No one dominates the other. No one imposes their law. But both canoes move side by side, in the same direction: the health of the living.

This working bond between our two peoples — those who were here before and those who came after — is not a fusion. It is a complementarity. Each brings what they know. Each learns from the other. The mycelium underground does not fuse roots — it connects them. That is exactly what this is.

You do not become Indigenous by joining this project. You remain who you are. But you learn to navigate side by side. You learn that your law is not superior. You learn that your language is not the only one. You learn that your spirituality is not the norm.

And you learn that it is possible.

VI. THE COMPLEMENTARITY OF DAILY LIFE

The Nabek farm does not produce for the market. It produces for the people. Meat, fibres, medicine, construction, energy — everything that comes from the land answers a need of first necessity. Not a desire for profit. Not speculation. Not export.

Surplus feeds the neighbour. Shortfall is filled by the neighbour. This is the economy of gift, not the economy of exchange. Hemp clothes. Beef feeds. Compost regenerates. Nothing is sold to the highest bidder. Everything is shared according to need.

This is not state communism. This is not green capitalism. It is something else. It is the law of the living applied to human economy.

VII. THE COMPLEMENTARITY TOWARD MOTHER EARTH

The mycelium is the invisible network that connects all lives underground. The Nabek farms are rivers of this network. Each farm is a node where the flow of the living passes, feeds, regenerates, and continues.

We do not own the river. We cross it. We do not own the mycelium. We tend it. We do not own the land. We return to it what colonization took from it: the freedom to be without an owner.

Every hectare returned to the mycelium is a hectare returned to the original law of this land. Every beef grazed in rotation is a step toward the end of extraction. Every hemp stalk is a thread of the Wampum reactivated.

VIII. WHAT WE ASK OF YOU

If you read this text and wish to join, here is what we ask of you:

1. You read this text to the end. Not diagonally. Not scanning. Word by word.
2. You read it a second time. At night, in silence. You see if it resonates.
3. You ask your questions. All of them. Even those that seem stupid. Even those that seem hostile.
4. You sign nothing. No contract. No legal commitment. Just a yes from the heart.
5. You come see. A farm. A caretaker. A piece of land. You set foot on it. You breathe. You listen.
6. And then you decide. Not for us. Not for the project. For yourself. For the land. For the generations to come.

IX. WHAT WE PROMISE

We do not promise wealth. We do not promise recognition. We do not promise it will be easy.

We promise this:

- That every land under this project will remain outside the market, outside taxes, outside ownership, forever.
- That every caretaker will be trained, accompanied, and never abandoned.
- That every decision will be made in agreement with the peoples of the territory.
- That work will be shared equitably — among peoples, among caretakers, among generations.
- That every surplus will go first to those who are hungry, not to those who pay the most.
- That the mycelium will always take priority over yield.
- That the Wampum will always take priority over profit.

X. CONCLUSION

40,000 farms. Not 40,000 properties. 40,000 entrusted responsibilities. 40,000 mycelium resistance nodes. 40,000 proofs that one can inhabit without possessing.

The land does not belong to humans. Humans belong to the land. And beneath the land, the mycelium waits for us. It has waited 400 years. It can wait longer. Or it can grow back — if we stop poisoning it, and if we stop pretending the land is ours.

This text is the doorway. You cross it, or you do not. Both choices are good. But there is no third way.

COMPLETE SYNTHETIC DOCUMENT

Regenerative Farm Quebec — Vision, Model, Value Chains and Projections

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SECTION I — VISION AND PHILOSOPHY

The Nabek project is born from a conviction: industrial agriculture is not a technique, it is a doctrine — the doctrine of discovery applied to the soil. The terra nullius projected underground. We arrive, we say: this land is empty, it is ours, we rename it, we cut it up, we poison it, we subjugate it.

The Nabek project proposes the opposite: 40,000 regenerative farms in Quebec, outside the government framework, without subsidy, anchored in the Two-Row Wampum of 1616. Each farm is a mycelium resistance node — a point where the underground fungal network is restored, where the soil becomes alive again.

The Two-Row Wampum (1616 Treaty)

Two canoes on a river. One carries the Europeans, the other the First Nations. Each keeps their own laws, customs, spirituality. No one dominates the other. The mycelium is the network of connection beneath both canoes — the invisible link that connects the visible (plants, animals, humans) and the invisible (mycelium, bacteria, minerals).

The Mycelium: the soil as brain

The mycelium is the underground fungal network — the nervous, immune and digestive system of the Earth. Natural density: 4.4 m of hyphae/cm³. Wild prairies: 6.6 m/cm³. Intensive farmland: 3.8 m/cm³ (~50% reduction). The holistic grazing of the Nabek project recreates the conditions of wild prairies at high fungal density.

KEY DATA — MYCELIUM RESTORATION

- 15 ha of pasture per farm × 0.15 m depth = 22,500 m³ of soil restored
- Density gain: 2.8 m of hyphae/cm³ (from 3.8 to 6.6)
- Hyphae restored per farm: 63 QUINTILLION metres
- At the scale of 40,000 farms: 2.52 sextillion metres of hyphae
- = 16,800 times the Earth-Sun distance

The 8 Nabek-Mycelium Junctions

- 1. Soil as brain** — 15 autonomous parcels = 15 neurons, emergent decision.
- 2. Soil as immune system** — Zero chemical inputs = natural immune system restored.
- 3. Soil as gut microbiome** — Compost (45 t/year) = food for the mycelium.
- 4. Collective intelligence** — 40,000 autonomous connected farms = human mycelium network.
- 5. Doctrine of discovery** — Refusal of the empire that subsidizes intensive agriculture.
- 6. Memory of the soil** — Traditional knowledge of First Peoples preserves the mycelium.
- 7. Water filtration** — 1 ha of mycelium-dense prairie filters ~10,000 m³ of water/year.
- 8. Ecological necrosis** — The current model destroys the mycelium; Nabek restores it.

SECTION II — NABEK FARM MODEL

Each Nabek farm is an autonomous 18-hectare unit integrating regenerative cattle farming and hemp rotation, with zero chemical inputs and a closed matter cycle.

BASE PARAMETERS PER FARM

- Total area: 18 hectares
- Holistic grazing: 15 ha (15 parcels of 1 ha in rotation)
- Hemp: 3 ha (4 strips of 0.75 ha)
- Cattle: 30 head (extensive grazing, 3-4 year cycle)
- Winter building: 400 m² hempcrete
- Composting: manure + straw (45 t/year), no slurry
- Pond + irrigation system: 1 unit per farm
- Mycorrhizal inoculation: *Glomus intraradices*, *Rhizophagus irregularis*, *Pisolithus tinctorius*

Annual regenerative cycle (8 steps)

- 1. Grazing** (May-October) — 30 cattle on 15 ha in rotation. Trampling = soil aeration. Urine = nitrogen for mycelium. Hyphae proliferate.
- 2. Manure + Straw** (November-April) — In the winter building, manure mixed with 30 t of straw. Soil mycelium (transported on hooves) colonizes the mix. Composting begins naturally.
- 3. Composting** (year-round) — Mycelium decomposes organic matter. Natural heat +15°C. Ammoniacal nitrogen transformed into stable organic nitrogen. No N₂O (no anaerobic fermentation).
- 4. Regenerated soil** (year-round) — Compost returned to soil. Organic matter increases. Soil structure improved. Microbial biodiversity restored.
- 5. Hemp** (April-September) — Roots up to 2 m deep. Oxygenation channels for hyphae. Mycorrhizal symbiosis. 30% faster growth with mycelium.
- 6. Harvest and processing** (August-October) — Hemp residues returned to soil (decomposition 3-6 months). Fibres and seeds processed locally.
- 7. Winter building** (November-April) — Porous hempcrete regulates humidity. Microclimate favourable to surrounding soil mycelium.
- 8. Return to soil** (year-round) — Compost feeds the earth. The mycelium distributes it throughout the underground network. The loop is closed — and the network expands.

Initial investment per farm

Investment item	Amount (CAD)
Winter building (hemcrete, 400 m ²)	280,000 \$
Fences and divisions (15 ha)	18,000 \$
Watering system	6,000 \$
Composting equipment	10,000 \$
Hemp harvest equipment (cooperative)	18,000 \$
Cattle purchase (30 head)	84,000 \$
Hemp seeds (3 ha)	1,080 \$
First year hay	11,664 \$
Year 1 labour (2 FTE)	96,000 \$
Miscellaneous year 1	18,000 \$
Soil prep + mycorrhizal inoculation	15,000 \$
TOTAL PER FARM	557,744 \$

Total project investment (40,000 farms): 22.31 billion \$ CAD

With cooperative strategy (equipment sharing 15-20 farms): 64-68% savings per farm (\$358,000-\$378,000 saved), revised investment of ~\$179,800-\$199,067 per farm.

SECTION III — BEEF VALUE CHAIN

40,000 farms × 30 cattle = 1,200,000 head. Annual cull rate: 30% (3-4 year cycle). 360,000 head to slaughter per year at maturity. Production: 311,535 tonnes of meat/year — Quebec self-sufficiency.

Required infrastructure

Infrastructure	Number	Capacity/year	Unit cost	Total cost
Artisanal slaughterhouses	144	2,500 head	2.5 M\$	360 M\$
Butcher shops	480	750 carcasses	500 k\$	240 M\$
Tanneries	240	1,500 hides	1.0 M\$	240 M\$
Charcuteries	206	1,750 carcasses	800 k\$	165 M\$
Valorization centres	533	75 farms	1.5 M\$	800 M\$
BEEF TOTAL	1,603	—	—	1.80 B\$

Jobs created by sector

Sector	Jobs	% of total
Livestock (farms)	80,000	71.0%
Transport / Logistics	20,000	17.7%
Administration / Coordination	4,000	3.5%
Valorization centres	3,200	2.8%
Butchery / Cutting	1,920	1.7%
Tannery / Leatherwork	1,680	1.5%
Slaughtering	864	0.8%
Charcuterie / Curing	1,029	0.9%
BEEF TOTAL	112,693	100%

Critical gaps

- **Tanneries:** 12 existing → 240 needed (shortage 228) — CRITICAL URGENCY. 130,000 lamb hides THROWN AWAY each year in Quebec.
- **Local slaughterhouses:** 24 existing → 144 needed (shortage 120) — MAJOR URGENCY.
- **Butcher shops:** ~80 existing → 480 needed (shortage 400).
- **Charcuteries:** ~20 existing → 206 needed (shortage 186).
- **Valorization centres:** ~0 existing → 533 needed — VIRTUALLY NON-EXISTENT.

Beef products and by-products — Zero waste

Meat — Deboned (tenderloin, ribeye, sirloin, etc.) + ground + offal (liver, kidneys, heart, tripe, blood).

Leather — Full grain (vegetable tanning: oak, chestnut, sumac) → shoes, leather goods, saddlery.

Bones — Broths, stocks, bone meal, gelatin, bone charcoal (water filter).

Fats — Lard, soap, candles, agricultural biofuel.

Blood — Black pudding, fertilizer, blood meal (12% nitrogen).

Horns and hooves — Buttons, combs, decorative objects, hoof powder (fertilizer).

Manure — 15 t/year composted with 30 t of straw → amendment for 15 ha of pasture.

BEEF CARBON BALANCE	
• CH ₄ emissions (cattle methane):	+84.0 t CO ₂ eq/farm/year
• N ₂ O emissions (soils):	+9.0 t CO ₂ eq/farm/year
• Soil sequestration (holistic grazing):	−11.5 t CO ₂ eq/farm/year
• Hemp sequestration (border):	−30.0 t CO ₂ eq/farm/year
• Compost carbon storage:	−5.0 t CO ₂ eq/farm/year
• NET BALANCE PER FARM:	−51.5 t CO ₂ eq/year (carbon negative from Year 1)

SECTION IV — HEMP TEXTILE VALUE CHAIN

Global hemp fibre market: \$3.67 B (2025) → \$24.65 B (2034), CAGR 24.14%. Textile/garment segment: \$1.06 B → \$7.38 B. Quebec imports 100% of its hemp textiles. No spinning, no weaving, no 100% Quebec hemp garment currently exists.

Base data

- Farms growing hemp: 4,000 (10% of 40,000 Nabek farms)
- Hectares per farm: 50 (hemp in rotation, not on the 3 ha border)
- Total hectares: 200,000
- Straw yield: 6 tonnes/ha → 1,200,000 t/year
- Raw fibre (30%): 360,000 t/year | Long textile fibre (50%): 180,000 t/year
- Hemp hurds (62.5%): 750,000 t/year | Dust: 60,000 t/year

Required infrastructure

Infrastructure	Number	Unit cost	Total cost
Industrial decorticators	96	3.0 M\$	288 M\$
Retting units	480	2.0 M\$	960 M\$
Hacklers	1,440	500 k\$	720 M\$
Spinning mills	3,600	1.5 M\$	5,400 M\$
Looms	4,800	800 k\$	3,840 M\$
Dyeing units	1,152	1.0 M\$	1,152 M\$
Garment units	4,800	600 k\$	2,880 M\$
Hurd processing	150	1.5 M\$	225 M\$
Cultivation equipment	4,000	50 k\$	200 M\$
TEXTILE TOTAL	16,518	—	15.66 B\$

Jobs created by sector

Sector	Jobs	% of total
Garment making	72,000	43.8%
Spinning	36,000	21.9%
Weaving	28,800	17.5%
Dyeing / Finishing	9,216	5.6%
Hemp cultivation	6,000	3.6%
Hackling	5,760	3.5%
Retting	3,840	2.3%
Transport / Logistics	1,200	0.7%
Hurd processing	750	0.5%
Decortication	480	0.3%
Administration	400	0.2%

TEXTILE TOTAL	164,446	100%
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Critical gaps

- **Spinning:** 0 existing → 3,600 needed — CRITICAL DEFICIT. Value-added loss: \$2 B/year.
- **Weaving:** 0 existing → 4,800 needed — CRITICAL DEFICIT.
- **Garment making:** 0 existing → 4,800 needed — CRITICAL DEFICIT.
- **Retting:** 0 existing → 480 needed — MAJOR DEFICIT.
- **Hackling:** 0 existing → 1,440 needed — MAJOR DEFICIT.
- **Decortication:** 1-2 existing → 96 needed (current capacity ~2,000 t/year vs 1,200,000 t/year).

Hemp fibre properties

Tensile strength: very high (superior to cotton and polyester). Breathability: excellent. Moisture absorption: 4× superior to cotton. Durability: very high. Biodegradability: 100%. Water needs: low (vs very high for cotton). Pesticides: none. CO₂ sequestration: high. Hemp yields 3-4× more fibre per hectare than cotton.

SECTION V — HEMP PHARMACOLOGICAL VALUE CHAIN

Industrial hemp (*Cannabis sativa* L.) has been legal in Canada since 1998 under Health Canada licence. CBD (cannabidiol) is the main non-psychoactive cannabinoid. Global CBD market: \$7.5 B USD in 2025, growth 15-20%/year. Quebec currently has NO supercritical CO₂ extraction company for pharmacological cannabinoids.

Base data

- Farms growing CBD hemp: 4,000 (same as textile, 10% of Nabek farms)
- Total hectares: 200,000 | Dried flower yield: 0.75 t/ha → 150,000 t/year
- Seed yield: 1.5 t/ha → 300,000 t/year | Straw/fibre yield: 6 t/ha → 1,200,000 t/year

Required infrastructure

Infrastructure	Number	Capacity/year	Unit cost	Total cost
CO ₂ extraction units	1,200	125 t flowers	2.0 M\$	2,400 M\$
Formulation units	900	25 M units	1.5 M\$	1,350 M\$
R&D; laboratories (ISO 17025)	24	—	5.0 M\$	120 M\$
Packaging lines	1,800	—	500 k\$	900 M\$
Oil presses	600	500 t seeds	300 k\$	180 M\$
Decorticators	96	12,500 t straw	3.0 M\$	288 M\$
Cultivation equipment	4,000	—	50 k\$	200 M\$
PHARMA TOTAL	7,590	—	—	5.44 B\$

Jobs created by sector

Sector	Jobs	% of total
Pharmaceutical formulation	10,800	26.1%
Packaging	10,800	26.1%
Cannabinoid extraction	9,600	23.2%
Hemp cultivation	6,000	14.5%
Seed oil pressing	1,800	4.3%
Transport / Logistics	1,200	2.9%
Fibre decortication	480	1.2%
R&D; laboratories	360	0.9%
Administration	400	1.0%
PHARMA TOTAL	41,440	100%

Critical gaps

- **CBD extraction:** 0 existing → 1,200 needed — CRITICAL DEFICIT. Value-added loss: \$500 M/year.
- **Pharmaceutical formulation:** 0 existing → 900 needed — CRITICAL DEFICIT.
- **ISO 17025 laboratories:** ~3 academic → 24 needed — MAJOR DEFICIT.

- **Hemp cultivation:** ~1,500 ha existing → 200,000 ha needed.
- **Seed oil presses:** 0 existing → 600 needed.
- **Decorticators:** 2 existing → 96 needed.

Pharmacological properties of CBD

Anticonvulsant action (Epidyolex® approved for rare epilepsy), anxiolytic effect, anti-inflammatory properties, analgesic action, neuroprotective effect. Interaction with serotonin 5-HT_{1A}, dopamine D₂, GABA receptors. Industrial hemp THC content: < 0.3% (vs > 0.3% for medical/recreational cannabis).

SECTION VI — HEMP PRODUCTS: CLEAN VS CONTAMINATED FIELDS

Hemp in the Nabek model is not a crop. It is a multi-purpose infrastructure that adapts to soil quality. The distinction rests on differential bioaccumulation:

DIFFERENTIAL BIOACCUMULATION
• ROOTS + LEAVES → HIGH concentration of heavy metals
• STEMS + FIBRES → LOW to MODERATE concentration
• SEEDS → VERY LOW concentration (except Cd, Ni)

Summary table of key differences

Criterion	Contaminated fields	Clean fields
Food entry	■ FORBIDDEN	■ ALLOWED
Direct skin contact	■ NOT RECOMMENDED	■ ALLOWED
Clothing textile	■ NOT RECOMMENDED	■ RECOMMENDED
Construction	■ Hempcrete (inerting)	■ Hempcrete (healthy housing)
Energy / Fuel	■ Bioethanol, etc.	■ Bioethanol, etc.
Green chemistry	■ Succinic acid, PHB	■ Food-grade bioplastics
Technical fibre	■ Composites, etc.	■ Textile, paper
Pharmacopoeia	■ FORBIDDEN	■ CBD, cannabinoids
Animal feed	■ FORBIDDEN	■ Seeds, cake
Cosmetics	■ FORBIDDEN	■ Oil, CBD, skincare

Products from CLEAN FIELDS (3 ha/farm)

Seeds (hempseed) — 2,400 kg/year: hulled (superfood omega-3), flour, milk, butter, protein powder, edible oil (cold-pressed), biodiesel B100/B20, glycerol.

Fibres (bast) — 7,800 kg/year: long fibre (clothing, linen), short fibre (stuffing), yarn, fabric, rope, twine, paper (recyclable 7-8 times), cardboard.

Hurds — 14,700 kg/year: hempcrete (walls, insulation), bricks, insulation panels ($R = 2.5-3.5 \text{ m}^2 \cdot \text{K/W}$), hemp wool, granules, mortar, fibre panels (furniture, biocomposite), bioplastics (PHA, PLA).

Leaves — 4,500 kg/year: CBD infusion (herbal teas), salad (young shoots), CBD extract (cosmetic), compost, mulch.

Roots — 2,000 kg/year: soil amendment, biochar (stable carbon >100 years).

Total biomass — 15,000 kg/year: biogas (anaerobic digestion), pellets (18 MJ/kg), farm compost.

Products from CONTAMINATED FIELDS (phytoremediation)

■■ GOLDEN RULE: No product enters the human or animal food chain.

Biofuels — Bioethanol (E85, E10), technical biodiesel, industrial biogas, heating pellets (industrial only).

Construction materials — Hempcrete (confinement of contaminants in mineral matrix), bricks, granulated insulation (no skin contact), fibre panels.

Industrial composites — Fibres for automotive, aerospace, rail. Industrial bioplastics. Furniture materials (not worn on skin).

Green chemistry — Succinic acid (biodegradable polymer precursor), PHB (bioplastic), phytomining (recovery Cu, Ni, Cd, Pb).

Technical textiles — Cordage, canvas, industrial nets. Geotextiles (soil stabilization, drainage).

Specialized processes — Phytomining (incineration 800-1000°C → metal recovery), phytostabilization (in situ immobilization), co-incineration in cement kiln (>1450°C → permanent immobilization in clinker).

HEMP ECONOMIC BALANCE (3 ha, clean fields)	
• Seeds (hempseed + oil):	\$3,600-4,800/year
• Textile fibres:	\$2,000-3,500/year
• Hurds (hempcrete + insulation):	\$1,500-2,500/year
• Leaves (infusion + compost):	\$500-1,000/year
• Biogas/pellets (energy):	\$800-1,200/year
• Pharmacopoeia (CBD, terpenes):	\$2,000-4,000/year (if regulation favourable)
• TOTAL POTENTIAL:	\$10,400-17,000/3ha/year Cost: \$3,500-5,000 Margin: \$6,900-12,000

SECTION VII — INTEGRATED CIRCULAR ECONOMY

The Nabek model aims for zero waste at the scale of each farm and the entire network. Every matter flow becomes a value flow. Nothing is lost, everything is transformed.

Closed circuits by value chain

BEEF VALUE CHAIN — Carcass cycle

Meat → human food | Bones → broths, meal, phosphates | Fats → soap, candles, biofuel | Hides → leather, leather goods, saddlery | Offal → pâtés, terrines, preserves | Blood → black pudding, fertilizer | Ashes → composting, soil amendment | Manure → compost, biogas

HEMP VALUE CHAIN — Plant cycle

Flowers → CBD, CBG, CBN extraction (pharma) | Seeds → edible oil, proteins, cake (food) | Long fibre → spinning, weaving, garment (textile) | Hurds → insulation, hempcrete, mulch, granules (construction) | Dust → plaster, granules | Roots → soil amendment | Leaves → compost, animal feed

CROSS-VALUE CHAIN SYNERGIES — The heart of the model

- Cattle manure → hemp field fertilization
- Hemp hurds → cattle bedding (absorbent, antibacterial)
- Hemp oil → cattle feed (omega-3, herd health)
- Hemp compost → pasture amendment (rich in potassium)
- Biogas (hemp residues + manure) → farm energy (heating, electricity)
- Hempcrete → winter building construction (hygrometric regulation)
- Hemp cake → cattle feed (30-35% protein)

ZERO WASTE — 100% VALORIZATION

- Beef: 100% of carcass valorized (meat, leather, bones, fats, blood, horns, manure)
- Hemp: 100% of biomass valorized (seeds, fibres, hurds, leaves, roots)
- No slurry (closed composting, no pit)
- No purchased fertilizers (compost + urine + manure + fungal inoculation)
- No energy bill (biogas produced on-farm)
- No antibiotics (natural medicine, strengthened immune system)

SECTION VIII — COOPERATIVE EQUIPMENT SHARING STRATEGY

The cooperative equipment sharing model among 15-20 farms reduces initial investment by 64-68% per farm (\$358,000-\$378,000 savings). 2,000 cooperatives cover the 40,000 farms.

Equipment pool composition (15-20 farms)

Equipment	Qty	Unit cost	Total cost	Share/farm (20)
John Deere S760 combine harvester	1	420,000 \$	420,000 \$	21,000 \$
11 m draper header (hemp)	1	85,000 \$	85,000 \$	4,250 \$
FiberCut mower (Bish)	1	25,000 \$	25,000 \$	1,250 \$
High-density baler	1	45,000 \$	45,000 \$	2,250 \$
FiberTrack 660 decorticator	1	150,000 \$	150,000 \$	7,500 \$
Mobile grain dryer (GSI)	1	35,000 \$	35,000 \$	1,750 \$
Flatbed trailer 8 m	2	8,000 \$	16,000 \$	800 \$
120 HP tractor (transport)	1	85,000 \$	85,000 \$	4,250 \$
4x4 pickup truck	1	55,000 \$	55,000 \$	2,750 \$
Tools and safety	1 lot	15,000 \$	15,000 \$	750 \$
Shared hangar (construction)	1	100,000 \$	100,000 \$	5,000 \$
COOP EQUIPMENT TOTAL			1,031,000 \$	51,550 \$

Total investment per farm (with cooperative)

- Shared equipment (cooperative share): \$51,550
- Individual equipment (75-90 HP tractor, mower, tedder, plough, seeder, sprayer, trailer): \$122,000
- Membership share: \$5,000
- Start-up fund: \$2,500
- **TOTAL PER FARM: \$179,800** (vs \$557,744 full purchase alone)
- **SAVINGS: \$377,944 per farm (68% savings)**

Cooperative governance structure

Legal form — Agricultural workers cooperative (C.T.A.) recognized in Quebec (Cooperatives Act).

General Assembly — Annual meeting (autumn, after harvest). Budget and harvest plan adoption. Board of directors election. Quorum: 50% + 1.

Board of Directors (5 elected members, 2-year term) — President (coordination), Vice-President (operations), Treasurer (finance), Secretary (documentation), Member (maintenance).

Harvest Committee (3 members + coordinator) — Daily planning, unforeseen management, yield tracking.

Membership — Share capital \$5,000. Minimum 3-year commitment. 12-month withdrawal notice. Early withdrawal penalty: 20% of share capital.

Harvest calendar and rotation

Week 1 (mid-September) — Preparation, maintenance, calibration, safety training.

Weeks 2-4 (late September) — Seed harvest: combine harvester, 3 ha/day, 5 farms/week, evening relocation.

Weeks 5-7 (early October) — Straw mowing: FiberCut, 3 ha/day, tedding included.

Week 8 (mid-October) — Baling: 2-3 bales/hour, storage under hangar.

Weeks 9-12 (October-November) — Decortication: FiberTrack 660, 2-3 farms/week.

ANNUAL OPERATING COSTS (cooperative of 20 farms)	
• Depreciation (10 years): \$5,780/farm	
• Insurance (2% value): \$1,156/farm	
• Maintenance (3% value): \$1,734/farm	
• Hangar storage: \$600/farm	
• Fuel: \$750/farm	
• Operator labour: \$1,750/farm	
• Training + admin + reserve: \$1,150/farm	
• TOTAL ANNUAL: \$12,920/farm (vs \$17,451 direct use)	

SECTION IX — INVESTMENTS AND 20-YEAR FINANCIAL PROJECTIONS

All amounts in Canadian dollars (CAD). No government subsidies. Personal project, without public funding. Real Quebec data, May 2026.

10-year staggered action plan

Year	New farms	Active	Invest.	Rev./farm	Cost/farm	Result
0	0	0	600.0 M\$	—	—	−600.0 M\$
1	2,000	2,000	1,115.5 M\$	66.3 k\$	93.2 k\$	−1,169.3 M\$
2	3,000	5,000	1,673.2 M\$	66.3 k\$	93.2 k\$	−1,807.8 M\$
3	4,000	9,000	2,231.0 M\$	107.7 k\$	95.3 k\$	−2,119.4 M\$
4	5,000	14,000	2,788.7 M\$	107.7 k\$	95.3 k\$	−2,615.1 M\$
5	6,000	20,000	3,346.5 M\$	140.9 k\$	97.0 k\$	−2,469.4 M\$
6	6,000	26,000	3,346.5 M\$	140.9 k\$	97.0 k\$	−2,206.3 M\$
7	5,000	31,000	2,788.7 M\$	165.8 k\$	98.3 k\$	−697.9 M\$
8 ★	4,000	35,000	2,231.0 M\$	165.8 k\$	98.3 k\$	+129.6 M\$
9	3,000	38,000	1,673.2 M\$	165.8 k\$	98.3 k\$	+889.7 M\$
10	2,000	40,000	1,115.5 M\$	165.8 k\$	98.3 k\$	+1,582.3 M\$

★ **BREAK-EVEN POINT: Year 8 (35,000 active farms)**

20-year overall balance

Indicator	Amount
Total investments	25.15 B\$
Cumulative revenues	120.86 B\$
Cumulative costs	83.84 B\$
NET RESULT 20 YEARS	+37.02 B\$
Overall ROI	147.2%
IRR (Internal Rate of Return)	11.76%
NPV (discount rate 5%)	+12.14 B\$
NPV (discount rate 8%)	+4.98 B\$
NPV (discount rate 10%)	+1.93 B\$

Key indicators per farm (at maturity, year 10)

Item	Amount
Total annual revenue per farm	165,800 \$
■ Hemp (3 ha)	75,534 \$ (45.6%)

■ ■ Cattle (30 head, 40% cull = 12 head)	90,228 \$ (54.4%)
Operating costs per farm	98,300 \$
■ ■ Labour (1.5 FTE)	72,000 \$ (73.2%)
■ ■ Supplementary feed	6,000 \$ (6.1%)
■ ■ Veterinary (natural medicine)	2,000 \$ (2.0%)
■ ■ Fuel and maintenance	3,500 \$ (3.6%)
■ ■ Insurance	2,500 \$ (2.5%)
■ ■ Building/fence maintenance	2,500 \$ (2.5%)
■ ■ Fungal inoculation (amortized)	3,250 \$ (3.3%)
■ ■ Cooperative fees (1.5%)	2,487 \$ (2.5%)
■ ■ Miscellaneous + seeds	4,080 \$ (4.1%)
Net margin per farm	67,400 \$/year
Return on investment	12.1%/year
Margin/revenue ratio	40.7%

SECTION X — CARBON BALANCE

The Nabek model is carbon negative from the first year. Composted manure (no slurry) eliminates N₂O emissions linked to anaerobic fermentation. Holistic rotational grazing captures carbon in the soil. Border hemp captures 30 t CO₂eq/year, largely offsetting cattle emissions.

Carbon balance per farm (t CO₂eq/year)

Element	Tonnes CO ₂ eq/farm/year	Type
CH ₄ (cattle methane)	+84.0	Emission
N ₂ O (soils)	+9.0	Emission
TOTAL EMISSIONS	+93.0	—
Soil sequestration (holistic grazing)	−11.5	Capture
Hemp sequestration (border, 3 ha)	−30.0	Capture
Compost carbon storage	−5.0	Capture
Fungal biomass (restored mycelium)	−17 to −28	Capture
TOTAL CAPTURE	−63.5 to −74.5	—
NET BALANCE PER FARM	−51.5 to −80.0	Carbon negative

At project scale (40,000 farms at maturity)

- Net balance per farm: −51.5 t CO₂eq/year (conservative, without mycelium)
- Project net balance: −2.06 Mt CO₂eq/year
- With restored mycelium: −68 to −80 t CO₂eq/farm/year → −2.72 to −3.2 Mt CO₂eq/year
- CO₂ captured by hemp (10-year growth): 1,200,000 tonnes
- Storage in hempcrete (>50 years): 588,000 tonnes
- Storage in biochar (roots): 120,000 tonnes
- Equivalent: 500,000 cars taken off the road (2.4 t CO₂/year/car)

WATER FILTRATION BY MYCELIUM

- 1 ha of mycelium-dense prairie filters ~10,000 m³ of water/year
- 15 ha × 10,000 m³ = 150,000 m³ of naturally filtered water per farm
- 40,000 farms = 6 BILLION m³ of filtered water/year
- = 1.6% of the St. Lawrence River flow (378 Bn m³/year)
- Without treatment plant — natural biological filtration

SECTION XI — 10-YEAR PHASING AND GOVERNANCE

Phase 1 — Years 1-2: Foundation and learning

Beef: 5 model slaughterhouses (1 per major region), 10 integrated butcher shops, 2 artisanal tanneries, 10 valorization centres.

Hemp Pharma: 20 pilot extraction units, 5 model laboratories, 50 formulation units, 100 oil presses, 10 decorticators, 500 pilot farms.

Hemp Textile: 5 pilot decorticators, 10 retting units, 20 hacklers, 50 spinning mills, 50 looms, 20 dyeing units, 50 garment units, 10 hurd processing units.

Phase 1 investment: ~\$4.5 B | **Jobs created:** ~35,000

Phase 2 — Years 3-5: Regional deployment

Beef: 50 additional slaughterhouses, 150 butcher shops, 50 tanneries, 200 valorization centres.

Hemp Pharma: 300 extraction units, 10 laboratories, 250 formulation units, 200 presses, 30 decorticators, 1,500 farms.

Hemp Textile: 30 decorticators, 150 retting units, 400 hacklers, 800 spinning mills, 1,000 looms, 300 dyeing units, 1,000 garment units, 50 hurd processing units.

Phase 2 investment: ~\$18 B | **Jobs created:** ~120,000

Phase 3 — Years 6-10: Full-scale deployment

Beef: 89 remaining slaughterhouses, 320 remaining butcher shops, 188 remaining tanneries, 323 remaining centres.

Hemp Pharma: 880 remaining extraction units, 9 remaining laboratories, 600 remaining formulation units, 300 remaining presses, 56 remaining decorticators, 2,000 remaining farms.

Hemp Textile: 61 remaining decorticators, 320 remaining retting units, 1,020 remaining hacklers, 2,750 remaining spinning mills, 3,750 remaining looms, 832 remaining dyeing units, 3,750 remaining garment units, 90 remaining hurd processing units.

Phase 3 investment: ~\$22.7 B | **Jobs created:** ~163,579

Global governance model

Level 1 — Central Nabek Cooperative

Elected Board of Directors (1 representative / 100 farms). Cross-sectoral committees. Common treasury. Share capital: \$15,000 per farm.

Level 2 — Regional Cooperatives (17 Quebec administrative regions)

Management of local infrastructure. Coordination between farmers and processors. Continuous training.

Level 3 — Sectoral Cooperatives

- Beef: C.T.A. per region
- Hemp Pharma: CPCHQ (Quebec Hemp Producers Cooperative)
- Hemp Textile: Quebec Hemp Textile Cooperative
- Agricultural equipment: 2,000 cooperatives of 15-20 farms

Priority training and succession

1. Regenerative cattle agronomy (MAPAQ, universities)
2. Hemp agronomy (CÉROM, Université Laval)
3. Butcher apprenticeship (revive the trade)

4. Tanner training (Écofaune boréale, CMCM)
5. Supercritical CO₂ extraction
6. GMP pharmaceutical formulation
7. Bast fibre spinning (France-Quebec transfer)
8. Hemp weaving
9. Ecological dyeing
10. Sustainable garment making (ESG UQAM, LaSalle College)

SECTION XII — DIRECTORY OF COMPANIES AND SUPPLIERS

QUEBEC HEMP COMPANIES

Company	Location	Activity	Contact
HECO Innovation Chanvre	Mirabel, QC	Hempcrete (lime-hemp), insulation	514 706-8186 / heco-innovation.com
Nature Fibres	Val-des-Sources, QC	Profib™ Mat insulation panels (NBC certified)	819 716-0141 / naturefibres.com
SQDPI	Sainte-Julie, QC	Pivot organization Quebec hemp value chain	Via MRC des Sources
Laboratoires Evogenics	Sainte-Julie, QC	R&D, decortication, technology transfer	514 666-3864 / laboratoireevogenics.com
Zone Agtech	L'Assomption, QC	Agricultural innovation hub, accelerator	450 938-5777 / zoneagtech.ca
TechniTextile Québec	Drummondville / Saint-Hyacinthe	Technical textiles materials cluster	450 418-5532 / technitextile.ca
VIVACO Coop	Victoriaville / Val-des-Sources	Cooperative group, owner of Nature Fibres	819 758-0671 / vivaco.coop
CÉROM	St-Mathieu-de-Beloeil, QC	Hemp varietal research	cerom.qc.ca
CATALIS Québec	Montréal, QC	Clinical trials coordination	catalisquebec.com
RPBQ	Québec, QC	Variety trials, production guides	rbpq.qc.ca

AGRICULTURAL EQUIPMENT SUPPLIERS (North America)

Supplier	Location	Specialty	Contact
Formation Ag Inc.	Monte Vista, Colorado, USA	Combines, FiberTrack decorticators	719 849-6633 / formation-ag.com
Bish Enterprises / Hemp Harvest Works	Giltner, Nebraska, USA	FiberCut mowers, cutter bars	bishenterprise.com
Henry's Hemp Harvester (HHH)	Burgdorf, Germany	Pharmaceutical-grade flower harvester	henryshempharvester.com
Triminor	Grass Valley, California, USA	Trimming, KIRPY CBD harvester	530 265-4277 / thetriminator.com
Panda Biotech	Dallas, Texas, USA	Large-scale decortication (10 t/h)	pandabiotech.com
Forever Green / Laumetris	Canada / Lithuania	KP-4 mower (European import)	getforevergreen.com

QUEBEC BEEF COMPANIES

Company	Location	Activity	Contact
Abattoir BL Inc.	Racine, Estrie	Federal cattle slaughtering	450 532-5444
Groupe ADEL / Viandes de l'Est	Luceville, BSL	Distribution, online shop	418 739-3544 / groupeadel.com
Montpak International	Laval, QC	Cull cows, deboning	montpak.com
Tannerie des Ruisseaux	Kamouraska, QC	Commercial tannery (only one in QC)	tanneriedesruisseaux.com

Forge & Cuir	Québec, QC	Artisanal work leather	forgeetcuir.com
Pascal le Boucher	Montréal (Villeray)	Short circuit, nose-to-tail	pascalleboucher.com
Boucherie du Terroir	Trois-Rivières, Sherbrooke	Highland, 60+ sausages	(819) 347-3209
Ferme Rustique	Ste-Croix-de-Lotbinière	Grass-fed beef, regenerative	fermerustique.ca
Ferme Saint-Vincent	St-Cuthbert, Lanaudière	100% certified organic	saint-vincentbio.com
Les Fermes Valens	Huntingdon, Montérégie	Organic and natural	fermesvalens.com

SUPPORT ORGANIZATIONS AND REGULATION

Health Canada — Controlled Substances Bureau (industrial hemp licences) | canada.ca/health-canada

MAPAQ — Agricultural support, slaughterhouse permits | mapaq.gouv.qc.ca

PBQ — Quebec Cattle Producers (certification, auctions) | pbq.qc.ca

CTAQ — Quebec Food Transformation Council | conseiltaq.com

CMCM — Quebec Leather Trades Council | cmcm.qc.ca

PGQ — Quebec Grain Producers | pgq.ca

CCME — Canadian Council of Ministers of the Environment (contaminated soil standards)

CFIA — Canadian Food Inspection Agency (food standards)

SECTION XIII — CONCLUSION

The Nabek project does not merely build farms. It plants a mycelium resistance network beneath the whole of Quebec. 40,000 farms = 40,000 resistance nodes. 720,000 hectares = 720,000 hectares of soil returned to life. 2.52 sextillion metres of restored hyphae = an underground network of biological resistance, invisible, indestructible.

This is not a metaphor. It is a living infrastructure that opposes the empire — underground, invisible, indestructible. The mycelium cannot be destroyed by decree. It cannot be banned by law. It cannot be bought by a multinational. It grows. It connects. It communicates. It resists.

Over 20 years, the project generates a net positive result of \$37.02 billion with an IRR of 11.76%. Break-even is reached in Year 8. At maturity, each farm produces \$165,800 in annual revenues for a net margin of \$67,400. The model is carbon negative from the first year (−51.5 t CO₂eq/farm/year) and creates 222,000 direct jobs.

311,535 tonnes of meat per year to feed Quebec.

600,000 tonnes of compost per year to regenerate the soil.

480,000 head slaughtered per year, each 100% valorized.

180,000 tonnes of textile fibre per year to clothe Quebec.

150,000 tonnes of CBD flowers per year to heal Quebec.

"We are not farmers. We are caretakers of the living. We do not produce meat. We restore a network. We do not build farms. We plant resistance."

The land does not belong to humans. Humans belong to the land. And beneath the land, the mycelium waits for us. It has waited 400 years. It can wait longer. Or it can grow back — if we stop poisoning it.

40,000 farms. 10 years. A choice.
Not just for the beef. For the hemp. For the bond. For the memory. For life.

*Developed by Nabek, in accord with the Kanienkehaka Longhouse
Personal project in Quebec | Without subsidy | Outside government framework
Real data | Quebec market prices (May 2026)
Unceded Land — Turtle Island — Year 410 of the 1616 Treaty
All amounts in Canadian dollars (CAD) — Exchange rate: 1 USD = 1.37 CAD
Complete document — June 2026*